

Hybrid Control Architecture for a Two-Wheeled Inverted Pendulum System

1 Introduction

This document describes a three-layer cascade hybrid control architecture developed to robustly stabilize a two-wheeled inverted pendulum system. The design integrates:

- **Outer-Loop NMPC (Nonlinear Model Predictive Control)** for predictive optimization and constraint handling,
- **Inner-Loop DSC (Dynamic Surface Control)** for fast, nonlinear compensation,
- **Adaptive NN (Neural Network) Augmentation** to learn and compensate for residual uncertainties.

The final control input is the sum of the NMPC, DSC, and NN outputs, with saturation applied to ensure that the command remains within practical limits.

2 System Model Overview

A linearized state-space model is used for the NMPC design:

$$\begin{aligned} A_{lin} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{bd}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{ad}{\Delta} & 0 \end{bmatrix}, & B_{lin} &= \begin{bmatrix} 0 \\ \frac{c}{\Delta} \\ 0 \\ -\frac{b}{\Delta} \end{bmatrix}, \\ C_{lin} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, & D_{lin} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \end{aligned}$$

with the parameters a , b , c , d , and Δ computed from the physical properties of the pendulum and wheels. This model is used to predict future states in the NMPC layer.

3 Control Architecture Layers

3.1 Outer-Loop NMPC

Purpose: The NMPC layer optimizes the control effort over a finite prediction horizon while ensuring that actuator constraints are respected.

Function:

- Uses the linearized state-space model to predict future system states.
- At every sampling step, an optimization problem is solved to minimize a quadratic cost function:

$$J = \sum_{k=1}^{N_{horizon}} \left(x_k^T Q_{nmpc} x_k + R_{nmpc} u_k^2 \right),$$

where Q_{nmpc} and R_{nmpc} are weighting matrices.

- The optimization is performed using an interior-point algorithm (via `fmincon`), and only the first control input from the optimal sequence is applied.

Tuning Parameters:

- **N_{horizon}:** The prediction horizon length. A longer horizon can yield better performance but increases computational complexity.
- **Q_{nmpc} and R_{nmpc}:** The state and control weighting matrices determine the trade-off between state regulation and control effort. Increasing Q_{nmpc} emphasizes state tracking, while increasing R_{nmpc} penalizes aggressive control inputs.

3.2 Inner-Loop DSC

Purpose: The DSC layer quickly compensates for the dominant nonlinearities in the system, particularly the deviations in the pendulum angle.

Function:

- Defines an error signal based on the pendulum angle $e = x_3$ (assuming the desired angle is zero).
- Uses a first-order filter to smooth the error derivative:

$$\dot{z} = -\lambda_{dsc} z + \dot{e},$$

where z is the filter state.

- Computes a corrective control action:

$$u_{dsc} = -\alpha_{dsc} (e + z),$$

where α_{dsc} is the DSC gain.

Tuning Parameters:

- λ_{dsc} : The filter coefficient that determines the speed of the error derivative response. Higher values lead to faster response but may amplify noise.
- α_{dsc} : The gain that scales the corrective action. A larger α_{dsc} accelerates error reduction but may result in excessive control effort if set too high.

3.3 Adaptive NN Augmentation

Purpose: The adaptive NN layer learns to compensate for residual uncertainties and unmodeled dynamics that the NMPC and DSC layers do not capture.

Function:

- A simple neural network is represented by a scalar weight w_{nn} that multiplies the error e :

$$u_{nn} = w_{nn} e.$$

- The weight w_{nn} is updated online using a gradient descent rule:

$$w_{nn} \leftarrow w_{nn} + \gamma_{nn} e^2,$$

where γ_{nn} is the learning rate.

- Anti-windup is enforced by clamping w_{nn} between predefined limits $w_{\min}$ and $w_{\max}$.

Tuning Parameters:

- γ_{nn} : The learning rate that determines the speed of adaptation. A lower value ensures smoother weight adjustments, whereas a higher value can lead to faster adaptation but may risk instability.
- $w_{\min}$ and $w_{\max}$: The clamping limits that prevent the NN weight from growing too large, ensuring robustness and stability.

3.4 Combined Control Input and Saturation

Function:

- The total control input is computed as the sum of the outputs of the three layers:

$$u_{\text{combined}} = u_{\text{nmPC}} + u_{\text{dsc}} + u_{nn}.$$

- A saturation function is applied to the combined control input to ensure it remains within practical limits:

$$u_{\text{total}} = \text{sat}(u_{\text{combined}}),$$

where the saturation limit u_{sat} restricts the maximum absolute control input.

Tuning Parameter:

- $\mathbf{u}_{sat}$: The saturation limit on the control input, which prevents the actuator from receiving commands that are too large.

4 Simulation and Implementation

Simulation Setup:

- A time step t_s and a final simulation time T_{final} are defined.
- The initial state is set with a small pendulum angle deviation (e.g., 0.1 rad) to test the controller's stabilization ability.

Main Simulation Loop: At each simulation step:

1. **NMPC Computation:**

- An optimization problem is solved over a prediction horizon $N_{horizon}$ using the linearized model.
- The cost function minimizes state deviations and control energy.
- Only the first control input from the optimized sequence is used as u_{nmpc} .

2. **DSC Update:**

- The error $e = x_3$ (pendulum angle) and its derivative are used to update the filter state z .
- The DSC control input is computed as $u_{dsc} = -\alpha_{dsc}(e + z)$.

3. **Adaptive NN Update:**

- The NN control input is calculated as $u_{nn} = w_{nn} e$.
- The NN weight is updated via gradient descent and clamped within $[w_{min}, w_{max}]$.

4. **Control Combination and Saturation:**

- The three control contributions are summed to form $u_{combined}$.
- The final control input is $u_{total} = \text{sat}(u_{combined})$, ensuring it does not exceed $\pm u_{sat}$.

5. **State Update:**

- The full nonlinear dynamics of the system are simulated (e.g., using Euler integration) to update the state vector.

5 Performance Evaluation and Tuning Guidelines

The multi-layered hybrid controller leverages the advantages of each component:

- **NMPC:** Ensures that the overall system performance is maintained and constraints are respected. Tuning the prediction horizon and weighting matrices is critical for balancing performance against computational load.
- **DSC:** Provides rapid compensation for nonlinearities. Adjusting λ_{dsc} and α_{dsc} directly impacts the speed and magnitude of the corrective action.
- **Adaptive NN:** Enhances robustness by learning residual errors. The learning rate γ_{nn} and clamping limits w_{min} and w_{max} must be tuned to allow for fast yet stable adaptation.
- **Saturation:** The saturation limit u_{sat} prevents excessive control commands, ensuring that the final input remains within practical bounds.

Iterative tuning of these parameters is typically required to achieve the desired balance between rapid stabilization, robustness, and minimal control effort.

6 Summary

This hybrid control architecture combines:

- **Outer-Loop NMPC** for predictive optimization and constraint management,
- **Inner-Loop DSC** for rapid compensation of nonlinear dynamics,
- **Adaptive NN Augmentation** to learn and compensate for residual uncertainties,
- and a **Saturation Mechanism** to ensure practical control effort.

Each layer has its own set of tuning parameters that affect the overall performance. By carefully tuning the NMPC horizon and weights, DSC gains, NN learning rate and clamping limits, and the saturation level, the controller achieves robust stabilization of the inverted pendulum system under varying operating conditions.